

IBVAP — Single-Page Technical Cheat Sheet

19. Quick Reference Technical Fact Sheet

Core Project Information

- **Project Name:** IBVAP (Intelligent Border Video Analytics Platform)
- **Domain:** AI Border Surveillance, Intrusion Detection & Cryptographic Cybersecurity
- **Hackathon:** Smart India Hackathon (SIH 2026) — Blockchain & Cybersecurity

Key Technical Parameters

Component	Technical Detail	Primary Source File
Object Detection Model	Ultralytics YOLO11 (yolo11s.pt / yolo11n.pt)	src/main.py
Detection Class	Class 0 (person class only)	src/main.py (<code>classes=[0]</code>)
Tracker Algorithm	ByteTrack (High/Low Conf Score Fusion)	bytetrack_day3.yaml
High Track Threshold	0.25 (<code>track_high_thresh</code>)	bytetrack_day3.yaml
Low Track Threshold	0.10 (<code>track_low_thresh</code>)	bytetrack_day3.yaml
New Track Threshold	0.30 (<code>new_track_thresh</code>)	bytetrack_day3.yaml
Track Buffer Memory	60 Frames (approx. 2s memory)	bytetrack_day3.yaml
Matching Threshold	0.80 (<code>match_thresh</code>)	bytetrack_day3.yaml
Database Tier 1	MongoDB Atlas (IBVAP.intrusion_events)	src/app.py
Database Tier 2 (Fallback)	Local CSV (output/intrusion_log.csv)	src/app.py
Database Tier 3 (Fallback)	Hardcoded Prototype Seed Data	src/app.py
Cryptographic Hash Algorithm	SHA-256 (hashlib.sha256)	src/app.py
Audit Root Hash File	output/audit_root_hash.txt	src/app.py
Blockchain Ledger File	output/blockchain_ledger.json	src/app.py
Ledger Architecture	Single-Node Local Hash-Linked Ledger (4 Blocks)	output/blockchain_ledger.json

Frontend Framework	Streamlit 1.38+ with Plotly & Custom CSS	<code>src/app.py</code> / <code>streamlit_app/</code>
Deployment Platform	Streamlit Community Cloud + GitHub	<code>streamlit_app/requirements.txt</code>

3 Key Metrics / Facts to Memorize

- **Root Hash:** `441e0cfff3badcfe7705ce8004b5f0189642d3033ccc370ded52b8e0cb9dcb05`
- **Verified Prototype Events:** 3 Events (`EVT-001` at frame 30, `EVT-002` at frame 108, `EVT-003` at frame 177).
- **Data Fallback Resilience:** MongoDB Atlas Cloud $\rightarrow$ Local CSV $\rightarrow$ Hardcoded Seed Data.

3 Key Phrases to Say to Judges

- *"IBVAP fuses real-time AI vision with cryptographic security auditability."*
- *"Our ByteTrack score fusion minimizes false alarms by keeping track IDs active across heavy occlusions."*
- *"We provide zero-downtime multi-tier fallback and cryptographic tamper evidence for border defense."*